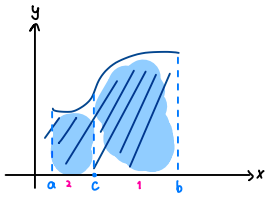
5

INTEGRALS

Rectangle $L \times W$

Triangle $\frac{1}{2} \cdot h \cdot b$

Circle πr^2



5.1

Areas and Distances

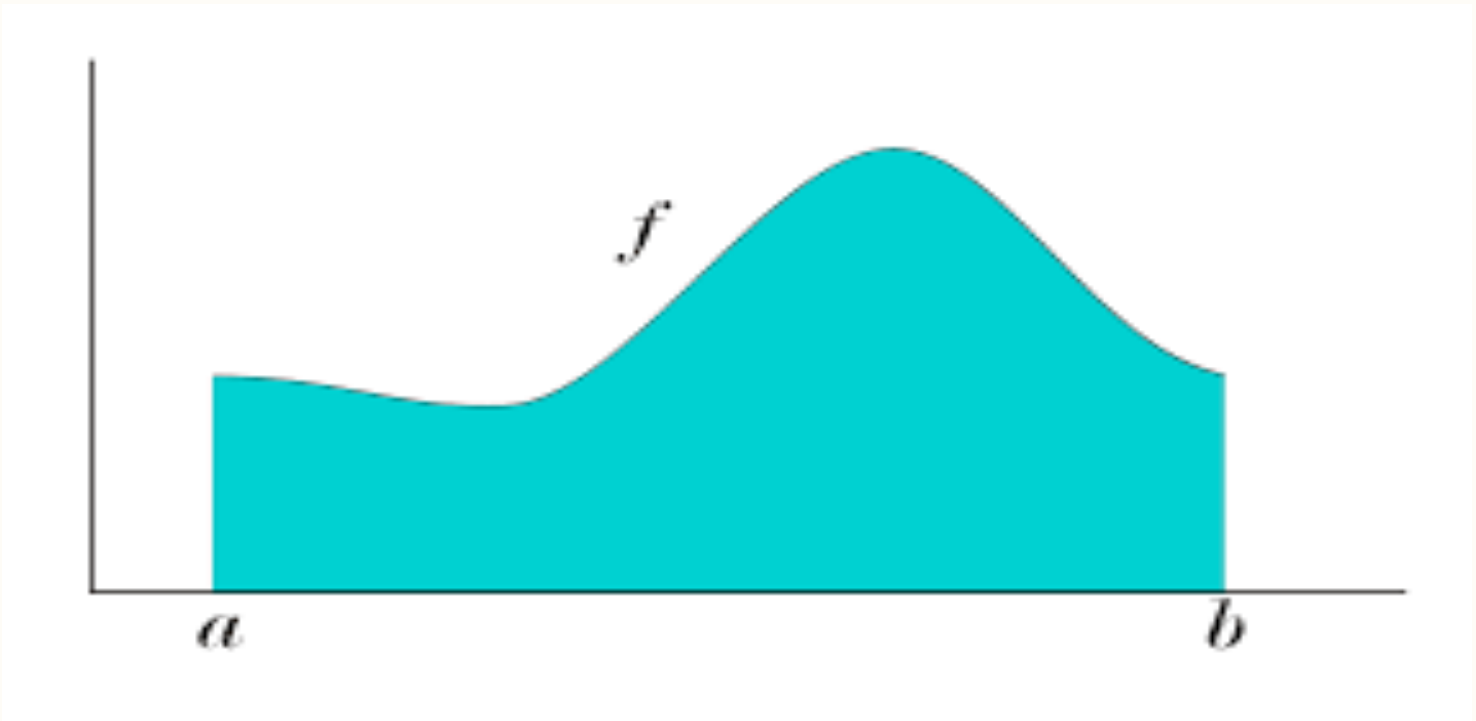
In this section, we will learn that:

We get the same special type of limit in trying to find the area under a curve or a distance traveled.

AREA PROBLEM

We begin by attempting to solve the area problem:

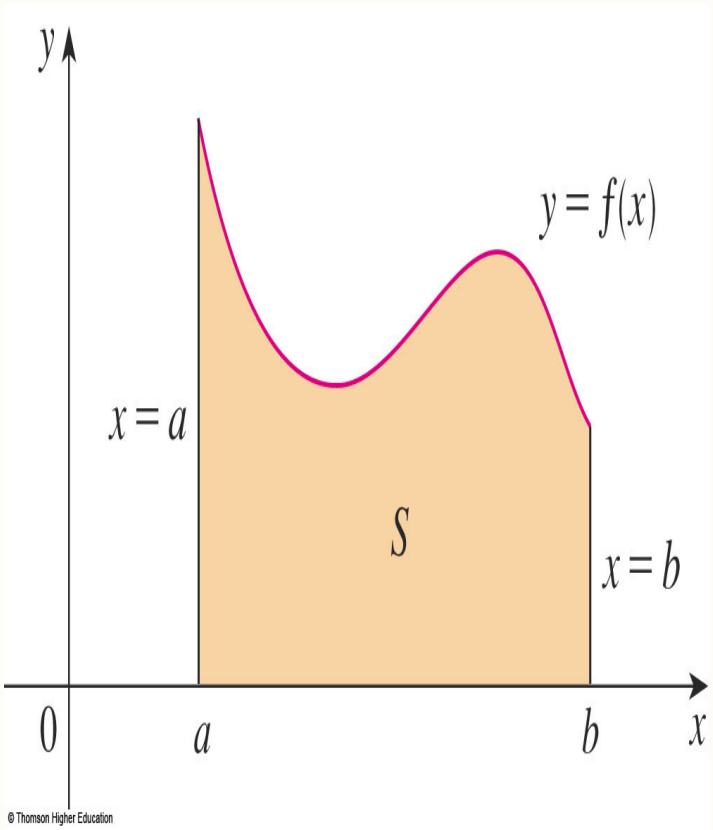
Find the area of the region S that lies under the curve $y = f(x)$ from a to b.



AREA PROBLEM

- This means that *S*, illustrated here, is bounded by:

- **The graph of a continuous function *f* [where $f(x) \geq 0$]**
- **The vertical lines $x = a$ and $x = b$**
- **The *x*-axis**



RECTANGLES

For a rectangle, the area is defined as:

The product of the length and the width



$$A = lw$$

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Example 1

Use 2 rectangles to estimate the area under the parabola $y = x^2$ from 0 to 1 (the parabolic region S illustrated in Figure 3).

$n = 2$ $[0, 1]$ $f(x) = x^2$

Figure 3

$w = \frac{b-a}{n}$

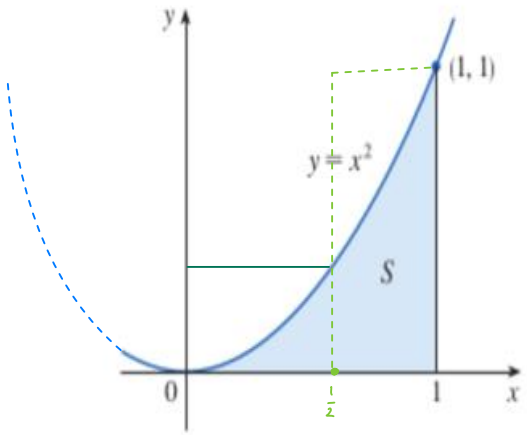
$w = \frac{1-0}{2} = \frac{1}{2}$

$A \simeq A_1 + A_2$

$= \frac{1}{2} f(1) + \frac{1}{2} f(\frac{1}{2})$

$= \frac{1}{2}(1)^2 + \frac{1}{2}(\frac{1}{2})^2$

$= \frac{1}{2} + \frac{1}{8} = \frac{5}{8} \rightarrow 0.625\ cm^3$



You split them into boxes and than you add them together

Me

Pick the first 2 points from the right

Me

Use 4 rectangles (n=4)

In the example given in class it she asked for 2 boxes. It can change the result depending on the amount of

Example 1

Use 5 rectangles to estimate the area under the parabola $y = x^2$ from 0 to 1 (the parabolic region S illustrated in Figure 3).

$n = 5$

$[0, 1]$

$f(x) = x^2$

Figure 3

$w = \frac{b-a}{n} \rightarrow \frac{1-0}{5} = 0.2$

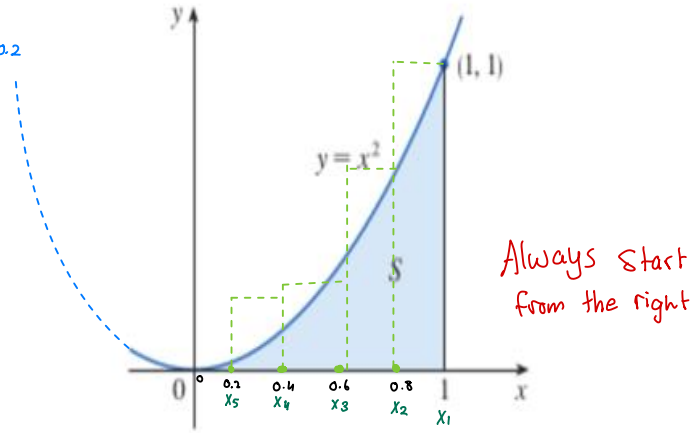
$A = A_1 + A_2 + A_3 + A_4 + A_5$

$(L \cdot w)$ for all of it

$= f(1)(0.2) + f(0.8)(0.2) + f(0.6)(0.2) + f(0.4)(0.2) + f(0.2)(0.2)$

$1^2 \cdot (0.2) + 0.8^2 \cdot (0.2) + 0.6^2 \cdot (0.2) + 0.4^2 \cdot (0.2) + 0.2^2 \cdot (0.2)$

$= 0.44 \text{ cm}^2$



Always start from the right

Can be written as $\sum f(x_i)\,\Delta x$

Make sure you connect all the columns to the point before it

Me

The more n increases we will get more of an accurate answer

Me

To get exact area let n—> ∞

Me $A = \lim_{n \to \infty} \sum f(x_i)\,\Delta x$

Use 4 rectangles (n=~~4~~ 5)

What we did last lecture

We estimated the area under a curve

$Y = f(X)$ $\quad X \in [a,b]$

Using n rectangles (right end points)

$$A \approx \sum_{i=1}^{n} f(x_i)\,\Delta x$$

$n \to m$

$$A = \lim_{n \to \infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$$

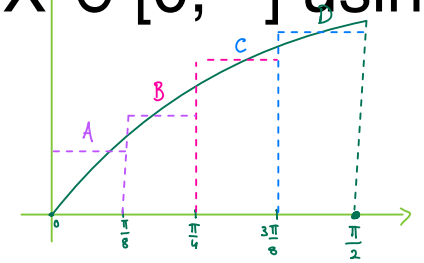
Removed sections

Left end points

Estimate the area under f (X)

= sin x

X € [0,] using 4 rectangles



$A \simeq f\left(\frac{\pi}{2}\right)\frac{\pi}{8} + f\left(\frac{3\pi}{8}\right)\frac{\pi}{8} + f\left(\frac{\pi}{4}\right)\frac{\pi}{8} + f\left(\frac{\pi}{8}\right)\frac{\pi}{8}$

$= \frac{\pi}{8}\left[f\left(\frac{\pi}{2}\right) + f\left(\frac{3\pi}{8}\right) + f\left(\frac{\pi}{4}\right) + f\left(\frac{\pi}{8}\right)\right]$

$= \frac{\pi}{8}\left[1 + 0.92 + 0.71 + 0.38\right]$

$\frac{\pi}{8}\left[3.01\right] = 1.18\,cm^2$

$\Delta x = w = \frac{b-a}{n}$

1) $w = \frac{\frac{\pi}{2} - 0}{4} \rightarrow w = \frac{\pi}{8}$

2) $\frac{\pi}{8} + \frac{\pi}{8} = \frac{\pi}{4}$

$\frac{\pi}{4} + \frac{\pi}{8} = \frac{3\pi}{8}$

$\frac{3\pi}{8} + \frac{\pi}{8} = \frac{\pi}{2}$

Do part 2 to get the points on the graph. Keep adding the w to itself.

Me

Remark:

1) $\lim_{n \to m} \frac{1}{n} = 0$

2) $\lim_{n \to \infty} c = c$

3) $\sum_{i=1}^{n} (a_i \pm b_i) = \sum_{i=1}^{n} a_i \pm \sum_{i=1}^{n} b_i$

4) $\sum_{i=1}^{n} c\, a_i = c \sum_{i=1}^{n} a_i$

5) $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

6) $\sum_{i=1}^{n} c = cn$

7) $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$

8) $\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$

Ex. for 5

L.H.S $\sum_{i=1}^{7} i$ $n = 7$

$1+2+3+4+5+6+7 = 28$

R.H.S $\frac{n(n+1)}{2}$ $\frac{7(8)}{2} = 28$

Ex. for 6

Choose $n = 8$ & $c = 3$

L.H.S $\sum_{i=1}^{8} 3 = 3+3+3+3+3+3+3+3 = 24$

R.H.S $cn = 3(8) = 24$

Example

5.

a. Estimate the area under the graph of $f(x) = 1 + x^2$ from $x = -1$ to $x = 2$ from to using three rectangles and right endpoints. Then improve your estimate by using six rectangles. Sketch the curve and the approximating rectangles.

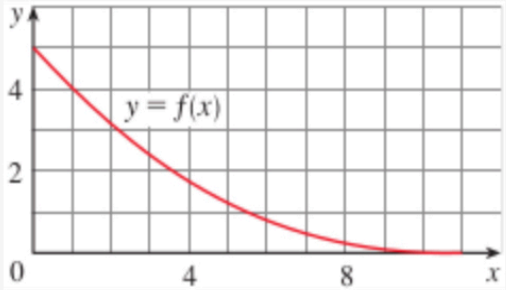
5.1 Exercises

1.

a. By reading values from the given graph of f, use five rectangles to find a lower estimate and an upper estimate for the area under the given graph of f from $x = 0$ to $x = 10$. In each case sketch the rectangles that you use.

Answer

b. Find new estimates using ten rectangles in each case.

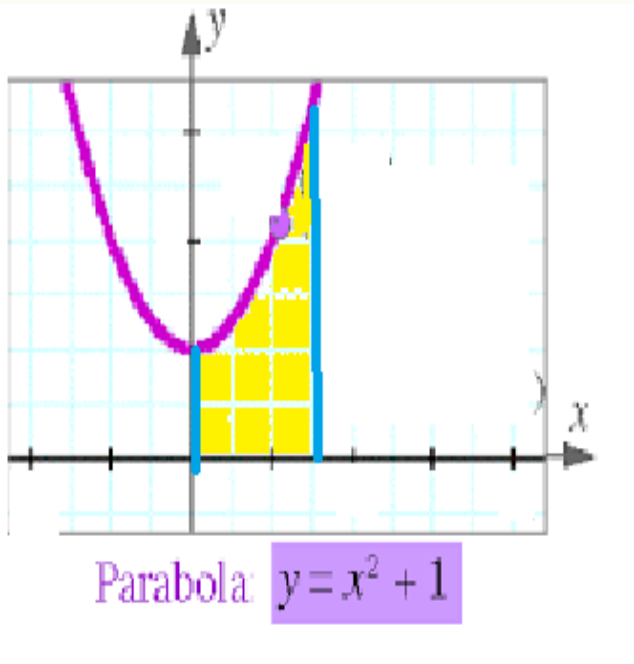


Example 1

– Use rectangles (n=4)to estimate the area under the parabola $y = x^2 + 1$

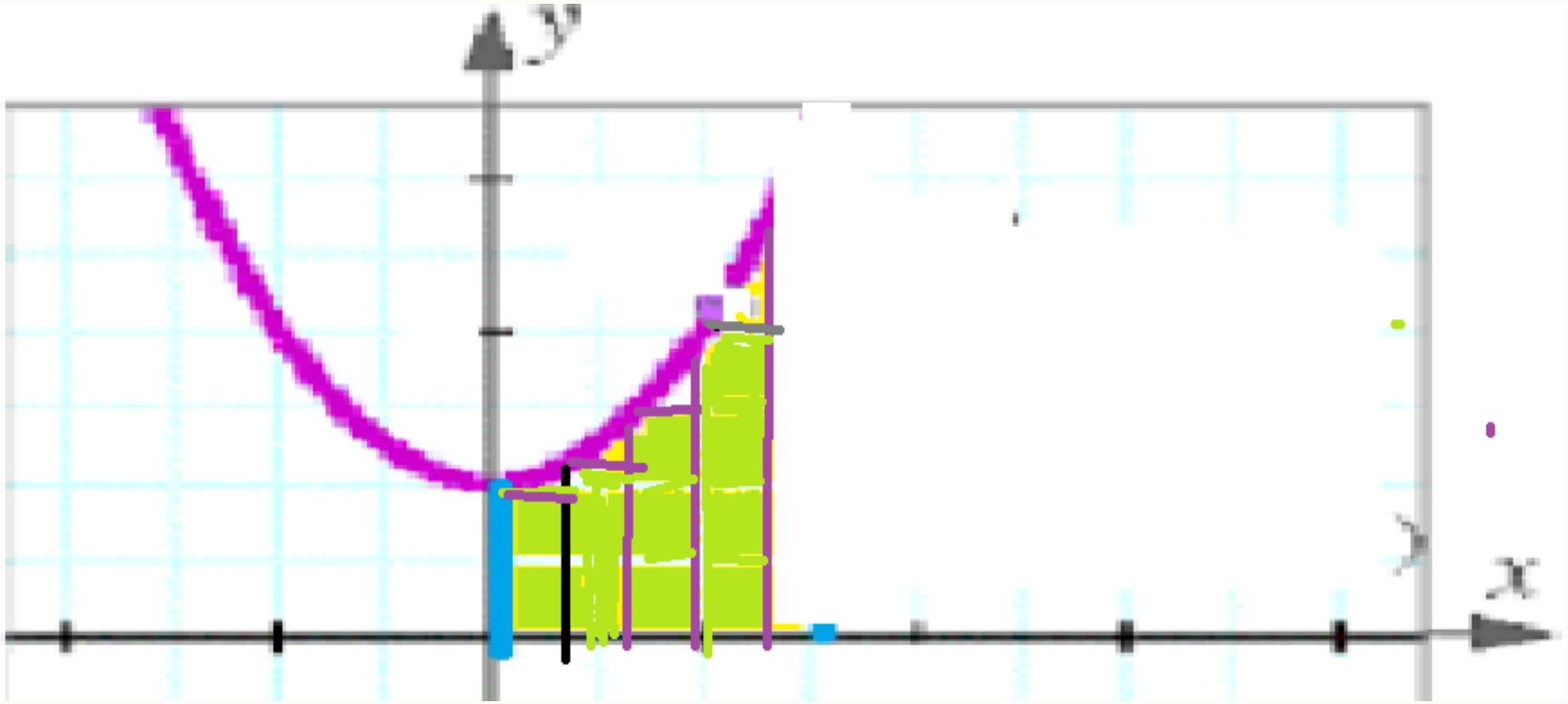
from 0 to 2.

Notice that the exact area A=4.67



Solution 1:

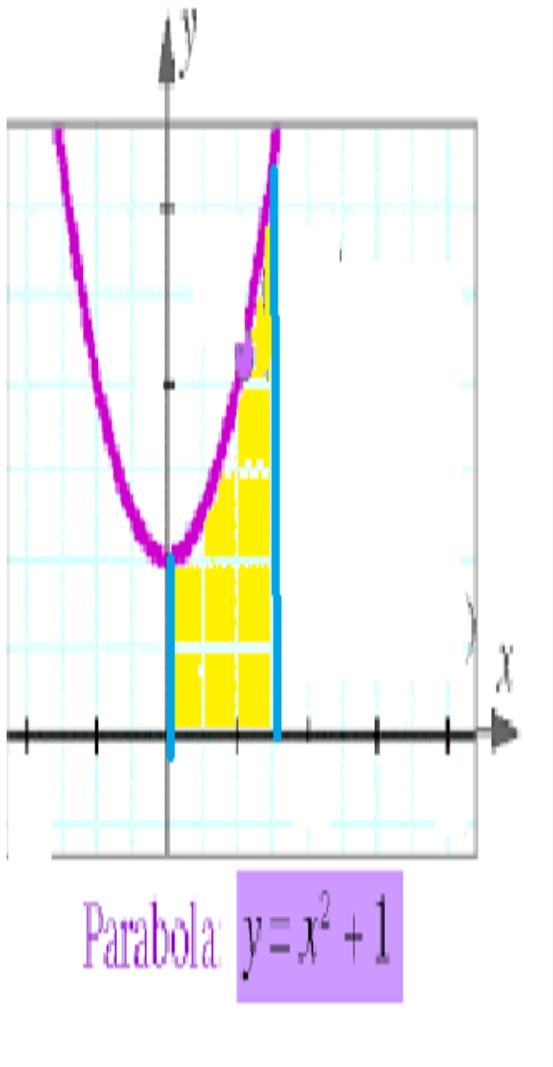
Suppose we divide *S* into four strips *S*1, *S*2, *S*3, and *S*4 by drawing the vertical lines starting from the left end points



Using the left end points :

– We first notice that:

$$\Delta x = \textit{the width of each rectangl} = \frac{2-0}{4} = 0.5$$



Notes that:

- The heights of these rectangles are the values of the function $f(x) = x^2 + 1$ at the left endpoints of the subintervals [0, 0.5],[0.5, 1], [1, 1.5], and [1.5, 2].
- $A_L = \sum_{i=1}^{4} \Delta x_i \, f(x_i)$=0.5(f(0)+f(0.5)+f(1)+f(1.5))

  =0.5(1+1.25+2+3.25)=3.75

Remark:

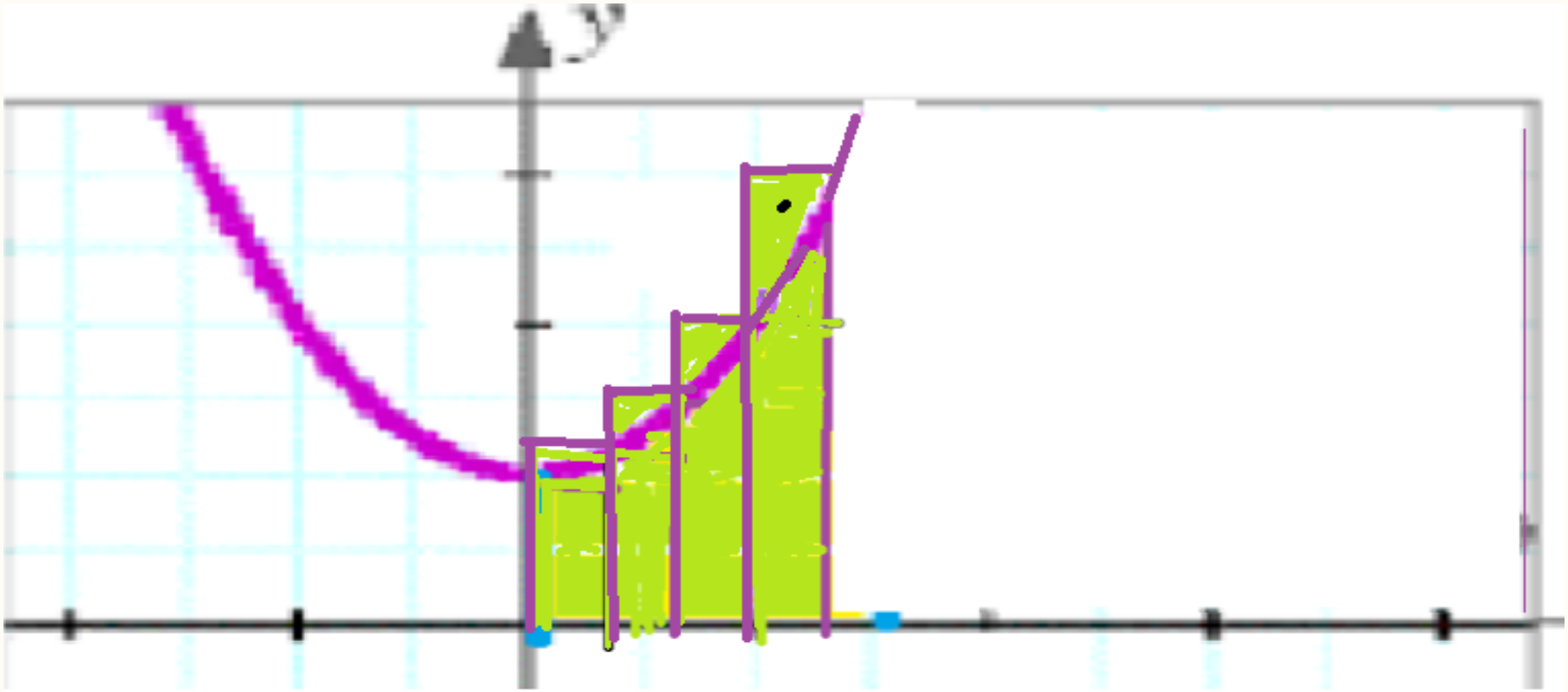
- We see the exact area $A = 4.67$ of S is larger than $AL = 3.75$

So, $AL < A$

Which mean :our approximation is under the accurate estimation.

Solution 2:

Suppose we divide *S* into four strips *S*1, *S*2, *S*3, and *S*4 by drawing the vertical lines starting from the right end points



Notes that:

- The heights of these rectangles are the values of the function $f(x) = x^2 + 1$ at the right endpoints of the subintervals [0, 0.5],[0.5, 1], [1, 1.5], and [1.5, 2].
- $A_R = \sum_{i=1}^{4} \Delta x_i \, f(x_i)$=0.5(f(0.5)+f(1)+f(1.5)+f(2))

  =0.5(1.25+2+3.25+5)=5.75

Remark:

- We see the exact area $A = 4.67$ of S is less than $AR = 5.75$

So, $AR > A$

Which mean :our approximation is above the accurate estimation.

Which mean:

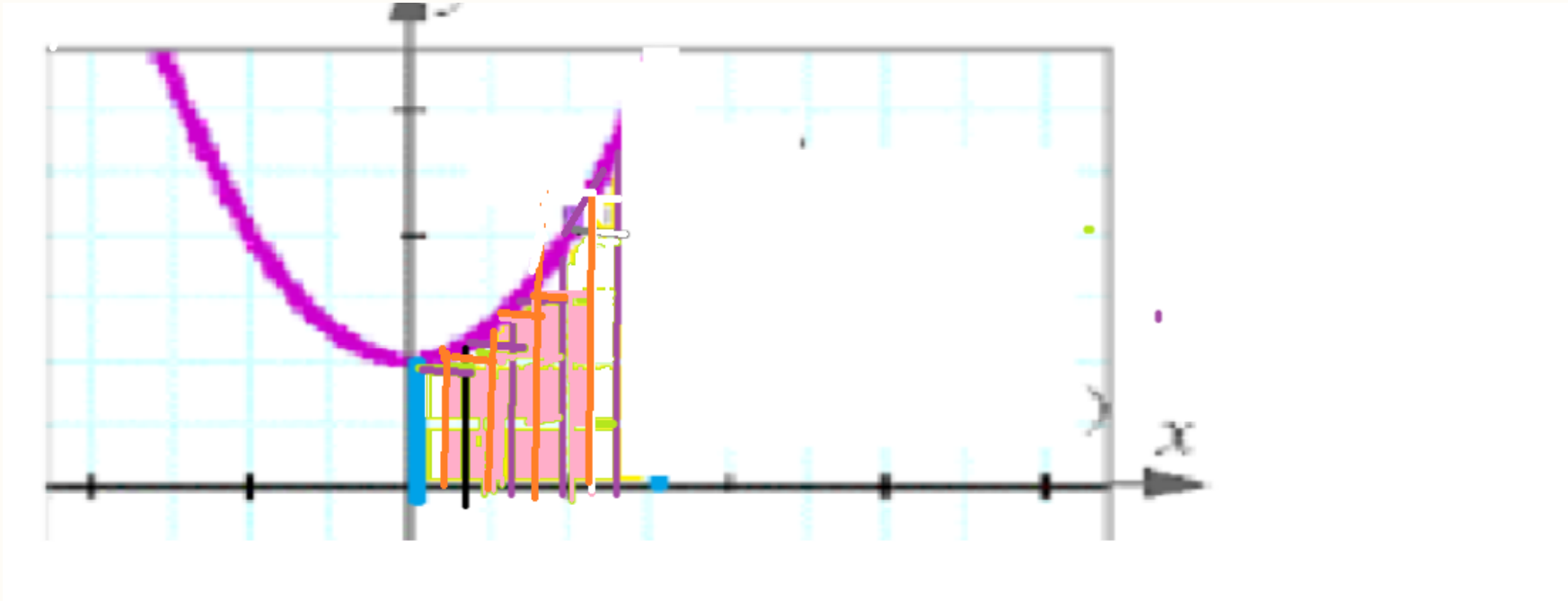
$$A_L < A < A_R$$

Moreover, some time if we use the average

$$\frac{A_L + A_R}{2} = \frac{3.25 + 5.75}{2} = 4.5 \cong 4.67 = exact\ area$$

Solution 3:

Suppose we divide *S* into four strips *S*1, *S*2, *S*3, and *S*4 by drawing the vertical lines starting from the midpoints



Notes that:

– The heights of these rectangles are the values of the function $f(x) = x^2 + 1$ at the midpoints of the subintervals [0, 0.5],[0.5, 1], [1, 1.5], and [1.5, 2].

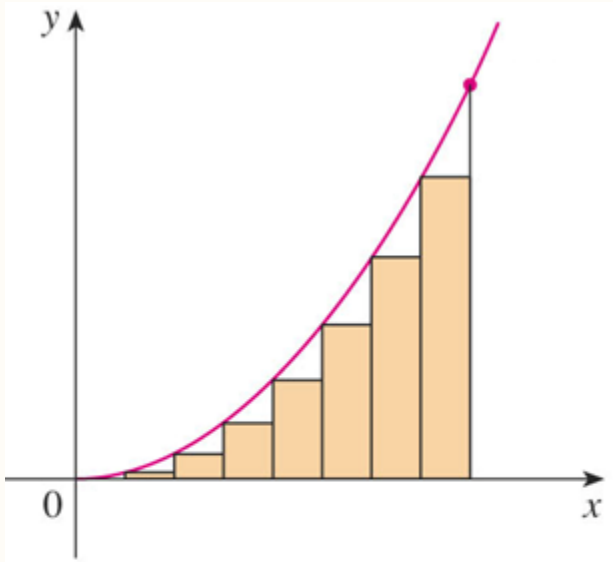
$$A_M = \sum_{i=1}^{4} \Delta x_i \, f(x_i)$$

$$=0.5(f(0.25)+f(0.75)+f(1.25)+f(1.75))$$

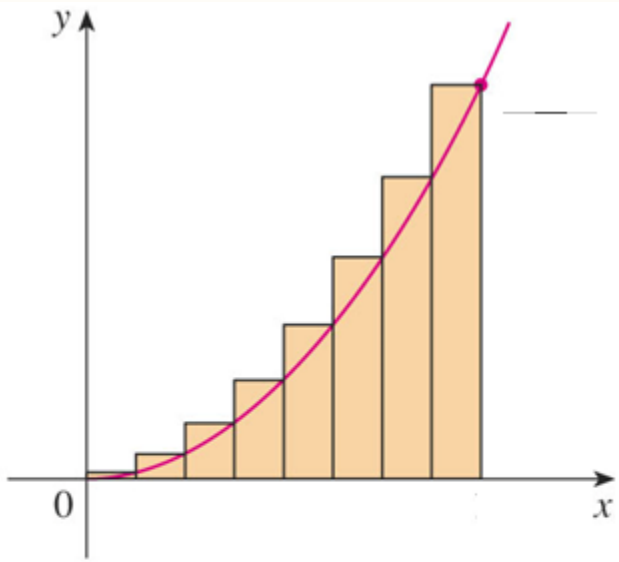
$$=0.5(1.25+2+3.25+5)=5.75$$



What do you think will happen if we repeat this procedure with a larger number of strips.

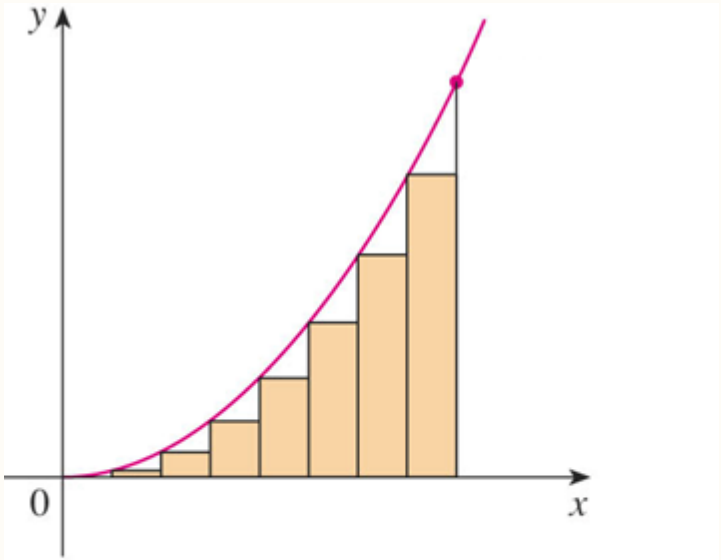


(a) Using left endpoints

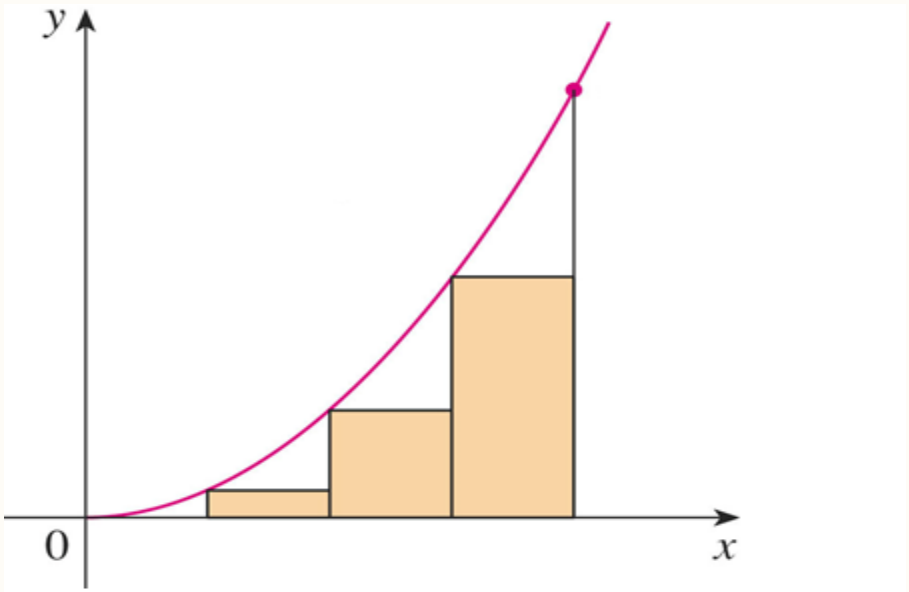


(b) Using right endpoints

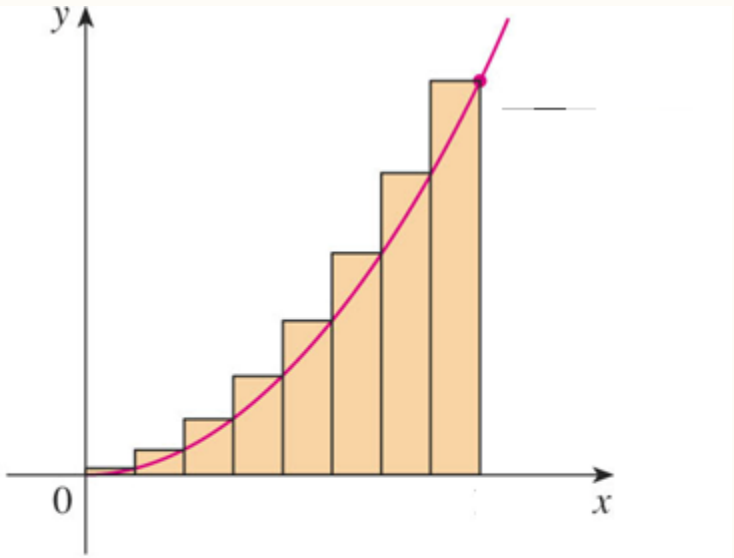
Using left end points



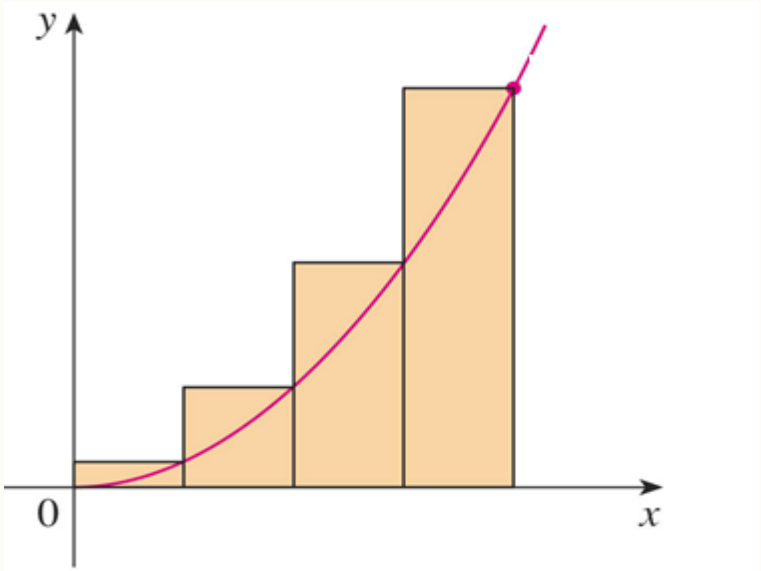
(a) Using left endpoints



Using right end points



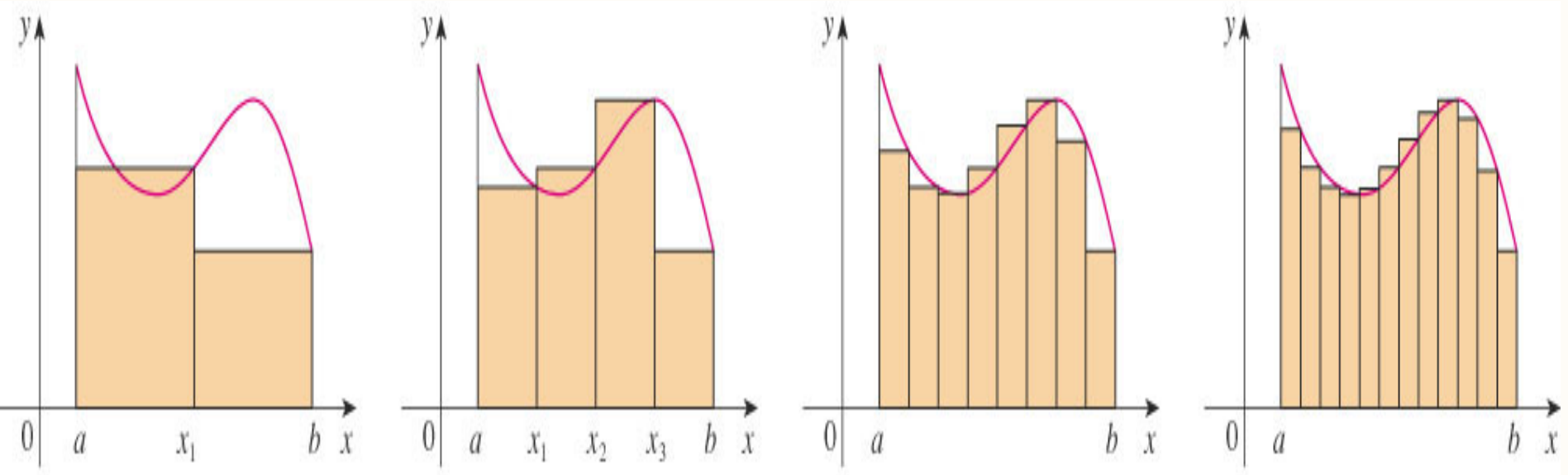
(b) Using right endpoints



Notice that

This approximation appears to become better and better as the number of strips increases,

that is, as $n \to \infty$.



(a) $n = 2$ (b) $n = 4$ (c) $n = 8$ (d) $n = 12$

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Definition

The area *A* of the region *S* that lies under the graph of the continuous function *f* is the limit of the sum of the areas of approximating rectangles:

$$A = \lim_{n\to\infty}[f(x_1)\Delta x + f(x_2)\Delta x + \ldots + f(x_n)\Delta x]$$

Remark:

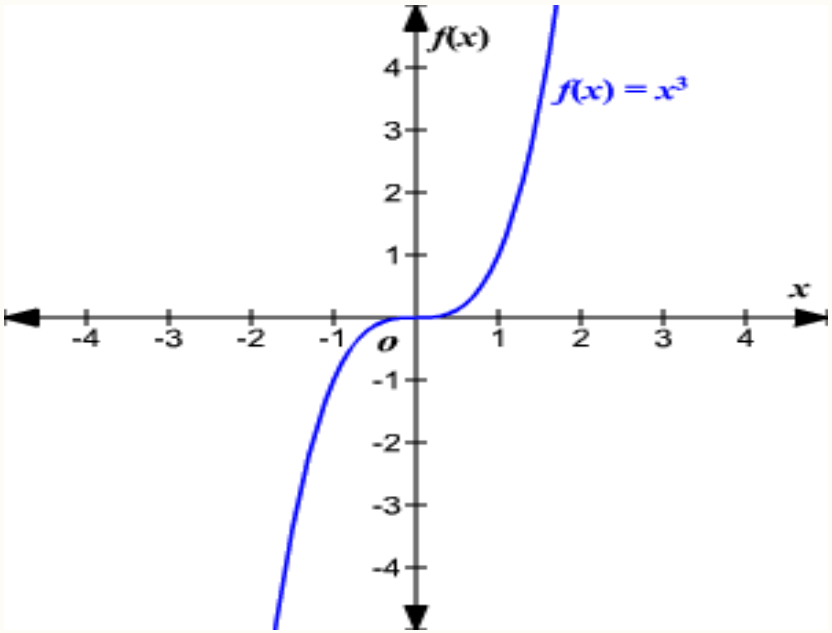
It can be proved that the limit in last Definition always exists—since we are assuming that f is continuous.

Example

– Use rectangles (n=4)to estimate the area under the curve $y = x^3$

from 0 to 4.

Notice that the exact area A=64



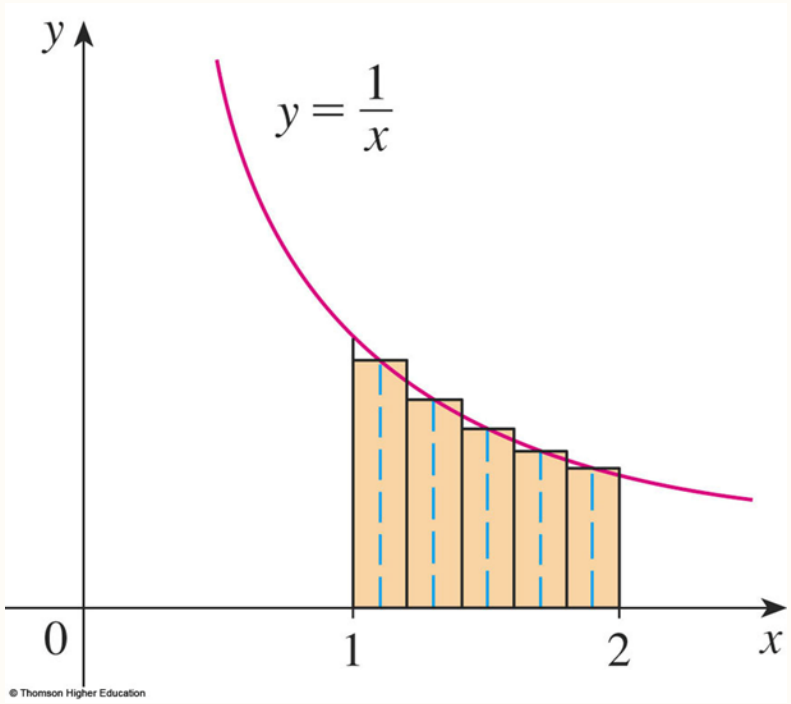
Solution

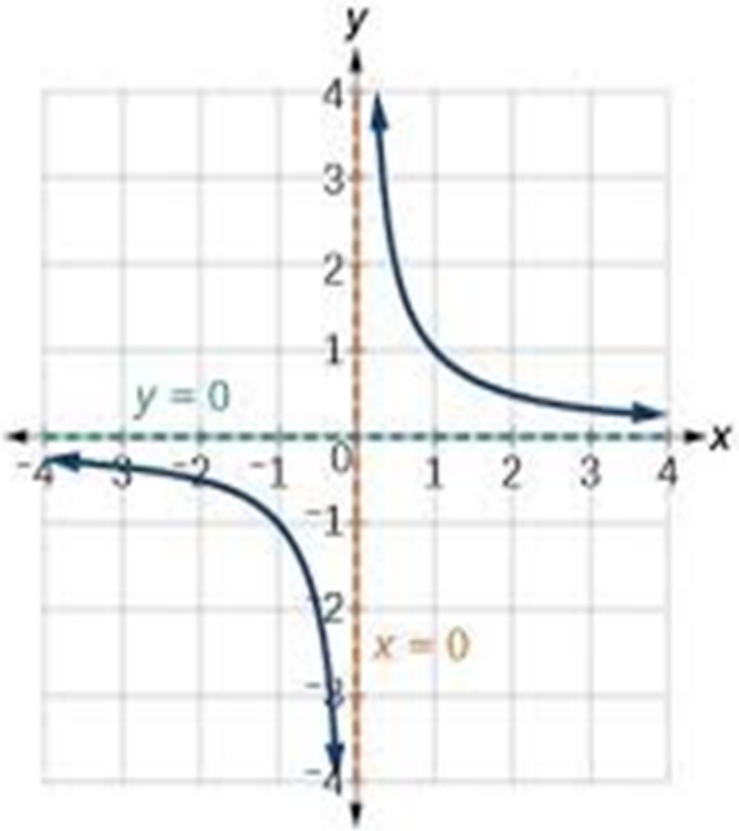
- $A_L = 36$
- $A_R = 100$
- Average=68
- $A_M = 62$
- Exact area=64

Example

– Use rectangles (n=4)to estimate the area under the curve $y = 1/x$ from 1 to 2.

Notice that the exact area A=0.6931





Solution

- $A_L = 0.76$
- $A_R = 0.635$
- Average=0.6975
- Exact area=0.6391

Remark

In last example we get

A*R < A <*AL



AREA PROBLEM

– We can also rewrite Formula 1 in the following way:

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

AREA PROBLEM

Example 3

– Let A be the area of the region that lies under the graph of $f(x) = e^{-x}$ between $x = 0$ and $x = 2$.

- Using right endpoints, find an expression for A as a limit. Do not evaluate the limit.

- Estimate the area by taking the sample points to be midpoints and using four subintervals and then ten subintervals.

Example 3 a

AREA PROBLEM

- Since $a = 0$ and $b = 2$, the width of a subinterval is:

$$\Delta x = \frac{2-0}{n} = \frac{2}{n}$$

- So, $x1 = 2/n$, $x2 = 4/n$, $x3 = 6/n$, $xi = 2i/n$, $x\text{n} = 2n/n$.

Example 3 a

– The sum of the areas of the approximating rectangles is:

$$R_n = f(x_1)\Delta x + f(x_2)\Delta x + \ldots + f(x_n)\Delta x$$

$$= e^{-x_1}\Delta x + e^{-x_2}\Delta x + \ldots + e^{-xn}\Delta x$$

$$= e^{-2/n}\left(\frac{2}{n}\right) + e^{-4/n}\left(\frac{2}{n}\right) + \ldots + e^{-2n/n}\left(\frac{2}{n}\right)$$

Example 3 a

AREA PROBLEM

– According to Definition 2, the area is:

$$A = \lim_{n\to\infty} R_n$$

$$= \lim_{n\to\infty} \frac{2}{n}\left(e^{-2/n} + e^{-4/n} + e^{-6/n} + \ldots + e^{-2n/n}\right)$$

– Using sigma notation, we could write:

$$A = \lim_{n\to\infty} \frac{2}{n} \sum_{i=1}^{n} e^{-2i/n}$$

AREA PROBLEM

Example 3 a

- It is difficult to evaluate this limit directly by hand.
- However, with the aid of a computer algebra system (CAS), it isn't hard.

- In Section 5.3, we will be able to find A more easily using a different method.

AREA PROBLEM

Example 3 b

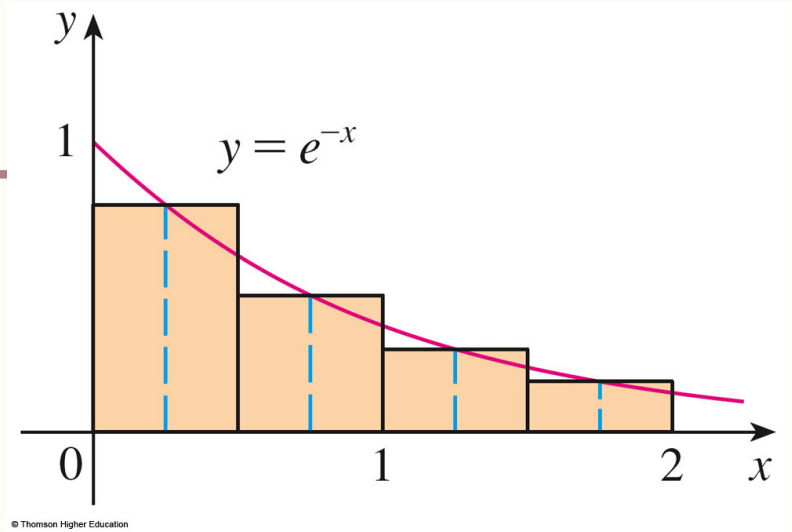
- With $n = 4$, the subintervals of equal width $\Delta x = 0.5$ are:
  [0, 0.5], [0.5, 1], [1, 1.5], [1.5, 2]

- The midpoints of these subintervals are:
  $x1^* = 0.25$, $x2^* = 0.75$, $x3^* = 1.25$, $x4^* = 1.75$

AREA PROBLEM

Example 3 b

– The sum of the areas of the four rectangles is:



$$M_4 = \sum_{i=1}^{4} f(x_i^*)\Delta x$$

$$= f(0.25)\Delta x + f(0.75)\Delta x + f(1.25)\Delta x + f(1.75)\Delta x$$

$$= e^{-0.25}(0.5) + e^{-0.75}(0.5) + e^{-1.25}(0.5) + e^{-1.75}(0.5)$$

$$= \tfrac{1}{2}(e^{-0.25} + e^{-0.75} + e^{-1.25} + e^{-1.75}) \approx 0.8557$$

AREA PROBLEM

Example 3 b

- With $n = 10$, the subintervals are:
  $[0, 0.2], [0.2, 0.4], \ldots, [1.8, 2]$
- The midpoints are:
  $x1^* = 0.1, x2^* = 0.3, x3^* = 0.5, \ldots, x10^* = 1.9$

AREA PROBLEM

Example 3 b

– Thus,

$$A \approx M_{10}$$

$$= f(0.1)\Delta x + f(0.3)\Delta x + f(0.5)\Delta x + \ldots + f(1.9)\Delta x$$

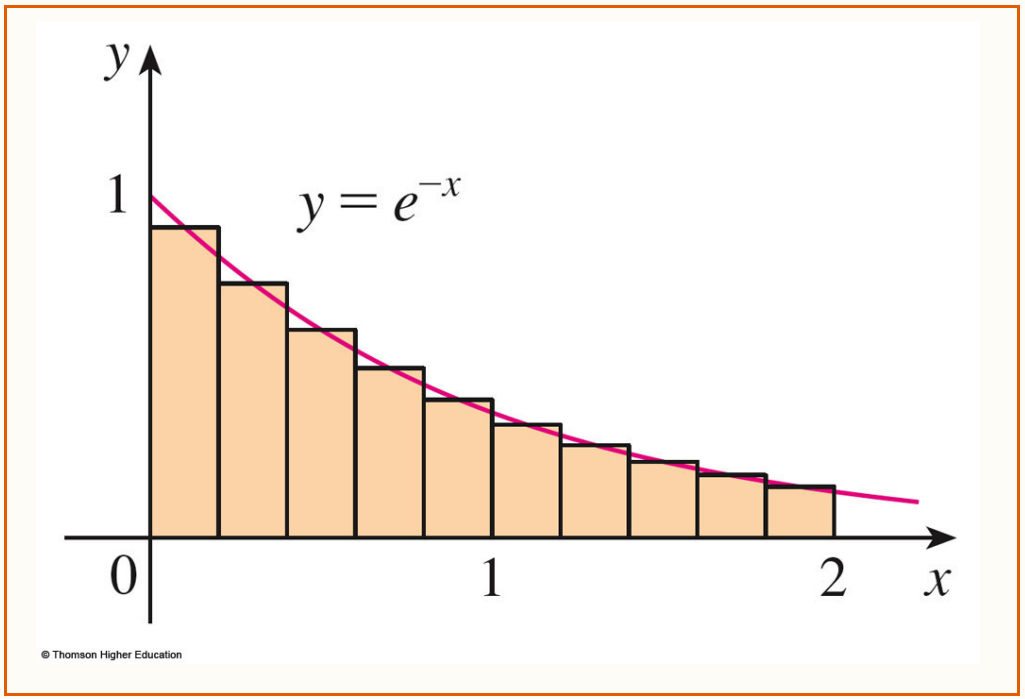
$$= 0.2(e^{-0.1} + e^{-0.3} + e^{-0.5} + \ldots + e^{-1.9})$$

$$\approx 0.8632$$

AREA PROBLEM

Example 3 b

- From the figure, it appears that this estimate is better than the estimate with $n = 4$.



DISTANCE PROBLEM

- Now, let's consider the distance problem.

- Find the distance traveled by an object during a certain time period if the velocity of the object is known at all times.

- In a sense, this is the inverse problem of the velocity problem that we discussed in Section 2.1

CONSTANT VELOCITY

- If the velocity remains constant, then the distance problem is easy to solve by means of the formula

  distance = velocity x time

VARYING VELOCITY

- However, if the velocity varies, it's not so easy to find the distance traveled.

- We investigate the problem in the following example.

Example 4

DISTANCE PROBLEM

– Suppose the odometer on our car is broken and we want to estimate the distance driven over a 30-second time interval.

DISTANCE PROBLEM

Example 4

– We take speedometer readings every five seconds and record them in this table.

Time (s)	0	5	10	15	20	25	30
Velocity (mi/h)	17	21	24	29	32	31	28

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DISTANCE PROBLEM

Example 4

– In order to have the time and the velocity in consistent units, let's convert the velocity readings to feet per second

(1 mi/h = 5280/3600 ft/s)

Time (s)	0	5	10	15	20	25	30
Velocity (mi/h)	17	21	24	29	32	31	28

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DISTANCE PROBLEM

Example 4

- During the first five seconds, the velocity doesn't change very much.

- So, we can estimate the distance traveled during that time by assuming that the velocity is constant.

Time (s)	0	5	10	15	20	25	30
Velocity (mi/h)	17	21	24	29	32	31	28

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Example 4

DISTANCE PROBLEM

– If we take the velocity during that time interval to be the initial velocity (25 ft/s), then we obtain the approximate distance traveled during the first five seconds:

25 ft/s x 5 s = 125 ft

Example 4

DISTANCE PROBLEM

- Similarly, during the second time interval, the velocity is approximately constant, and we take it to be the velocity when $t = 5$ s.

- So, our estimate for the distance traveled from $t = 5$ s to $t = 10$ s is:

$$31 \text{ ft/s} \times 5 \text{ s} = 155 \text{ ft}$$

DISTANCE PROBLEM

Example 4

– If we add similar estimates for the other time intervals, we obtain an estimate for the total distance traveled:

$$(25 \times 5) + (31 \times 5) + (35 \times 5)$$
$$+ (43 \times 5) + (47 \times 5) + (46 \times 5)$$
$$= 1135 \text{ ft}$$

Example 4

DISTANCE PROBLEM

- We could just as well have used the velocity at the end of each time period instead of the velocity at the beginning as our assumed constant velocity.

- Then, our estimate becomes:

$$(31 \times 5) + (35 \times 5) + (43 \times 5) + (47 \times 5) + (46 \times 5) + (41 \times 5) = 1215 \text{ ft}$$

Example 4

DISTANCE PROBLEM

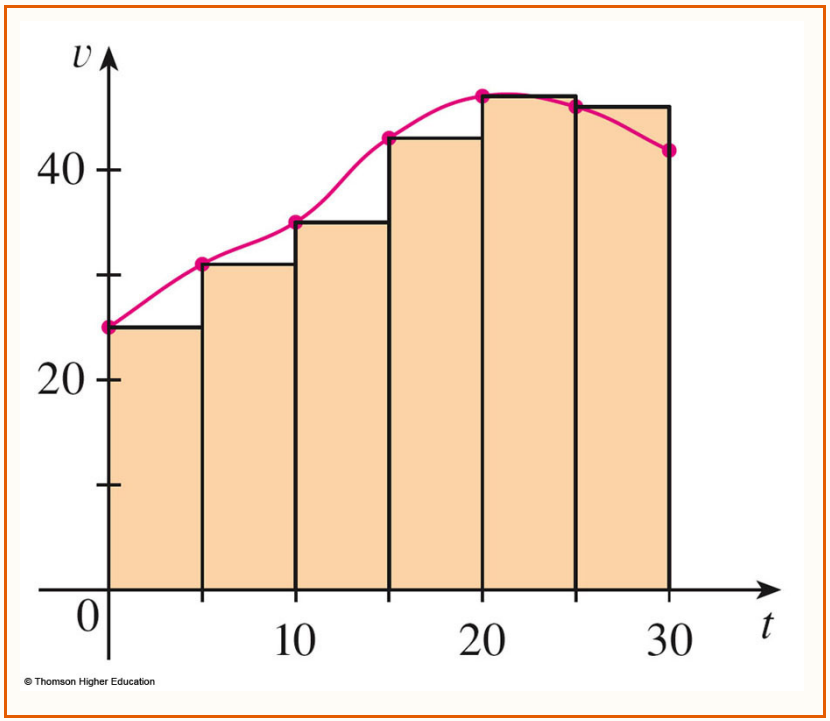
– If we had wanted a more accurate estimate, we could have taken velocity readings every two seconds, or even every second.

DISTANCE PROBLEM

- Perhaps the calculations in Example 4 remind you of the sums we used earlier to estimate areas.

DISTANCE PROBLEM

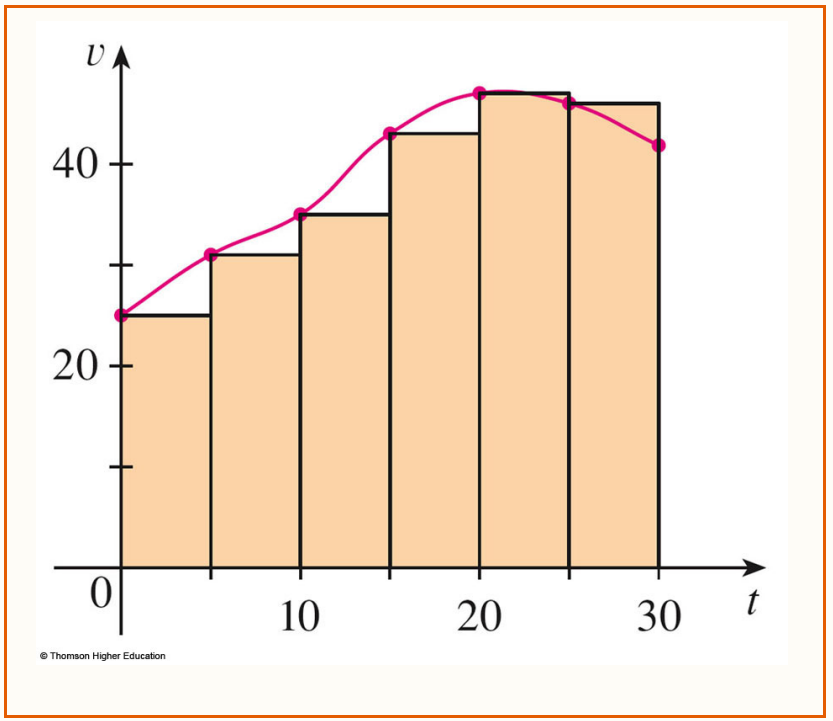
– The similarity is explained when we sketch a graph of the velocity function of the car and draw rectangles whose heights are the initial velocities for each time interval.



DISTANCE PROBLEM

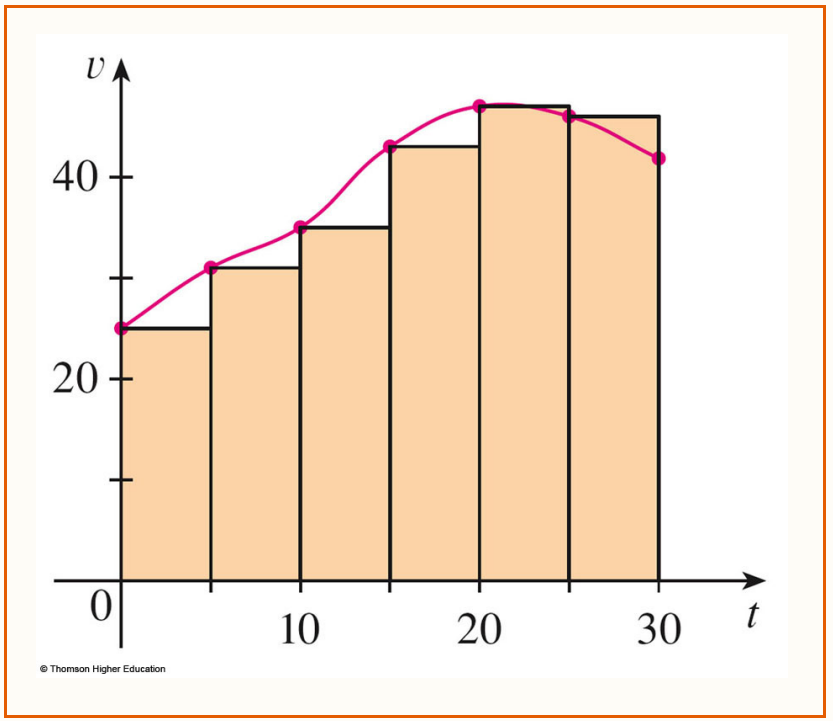
- The area of the first rectangle is 25 x 5 = 125, which is also our estimate for the distance traveled in the first five seconds.

- In fact, the area of each rectangle can be interpreted as a distance, because the height represents velocity and the width represents time.



DISTANCE PROBLEM

– The sum of the areas of the rectangles is $L6 = 1135$, which is our initial estimate for the total distance traveled.



DISTANCE PROBLEM

– In general, suppose an object moves with velocity

$$v = f(t)$$

where $a \leq t \leq b$ and $f(t) \geq 0$.

– So, the object always moves in the positive direction.

DISTANCE PROBLEM

– We take velocity readings at times

$t0(= a), t1, t2, \ldots, tn(= b)$

so that the velocity is approximately constant on each subinterval.

– If these times are equally spaced, then the time between consecutive readings is:

$$\Delta t = (b - a)/n$$

DISTANCE PROBLEM

- During the first time interval, the velocity is approximately $f(t0)$.

- Hence, the distance traveled is approximately $f(t0)\Delta t$.

DISTANCE PROBLEM

– Similarly, the distance traveled during the second time interval is about $f(t1)\Delta t$ and the total distance traveled during the time interval $[a, b]$ is approximately

$$f(t_0)\Delta t + f(t_1)\Delta t + \ldots + f(t_{n-1})\Delta t$$

$$= \sum_{i=1}^{n} f(t_{i-1})\Delta t$$

DISTANCE PROBLEM

– If we use the velocity at right endpoints instead of left endpoints, our estimate for
the total distance becomes:

$$f(t_1)\Delta t + f(t_2)\Delta t + \ldots + f(t_n)\Delta t$$

$$= \sum_{i=1}^{n} f(t_i)\Delta t$$

DISTANCE PROBLEM

- The more frequently we measure the velocity, the more accurate our estimates become.

DISTANCE PROBLEM

Equation 5

- So, it seems plausible that the exact distance d traveled is the limit of such expressions:

$$d = \lim_{n\to\infty} \sum_{i=1}^{n} f(t_{i-1})\Delta t = \lim_{n\to\infty} \sum_{i=1}^{n} f(t_i)\Delta t$$

- We will see in Section 5.4 that this is indeed true.

SUMMARY

- Equation 5 has the same form as our expressions for area in Equations 2 and 3.

- So, it follows that the distance traveled is equal to the area under the graph of the velocity function.

SUMMARY

- In Chapters 6 and 8, we will see that other quantities of interest in the natural and social sciences can also be interpreted as the area under a curve.
- Examples include:

- Work done by a variable force
- Cardiac output of the heart

SUMMARY

– So, when we compute areas in this chapter, bear in mind that they can be interpreted in a variety of practical ways.